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Minor Research + implementation project flow.

Step 1: Formulating objective function
- Clearly defines what problem you are solving and what Success looks like.
  output should be:-
  - 1 main objective
  - 2-4 sub - objective

Step 2: Extensive Literature survey
where to search:
- IEEE xplore
- Springer (Nature springer)
- Elsevier
- Taylor & Francis

output table:

Author	Year	method used	Dataset	Accuracy	Limitation

Step 3: Developing a Hypothesis
A hypothesis is a testable assumption
ex:
H_0 (Null hypothesis)
H_1 (Alternate hypothesis)

Step 4: Research formulation
This is where you design your approach
Include:
- Problem definition
  - Input → process → output
  - methodology flow

Add a block diagram/ flow chart

Step 5: Data Collection

- kaggle
- UCI ML Repository
- MPDI / NHMC
- Data.gov.in
- NIH (National institute of health)

Step 6: Data Analysis

Here you explore the data.

Do this:

- Handle missing values
- Remove duplicates
- Normalize / encode data
- use graphs

Step 7: Execution of the project

This is the actual build phase.

Includes:-

- Algorithm implementation
- Model training
- System development (if app / web involved)

Step 8: Analyze the Data (Post-Execution).

Now analyze the result, not raw data.

Includes:-

- Accuracy, precision, recall
- confusion matrix
- comparison between models

Create tables & graph - very scoring-friendly.

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Page ______

Step 9: Hypothesis Testing

Match results with your hypothesis

ex:-

Since the proposed model achieved an accuracy of 92%, the null hypothesis is rejected and the alternate hypothesis is accepted.

Step 10: Generalization & Interpretation

Explain what the result mean in real life.

ex:-

- Can this model work on other datasets?
- can it be used in hospitals?
- what are the constraints?

Also mention future scope.

Step 11: Documentation & presentation

↳ merits of the project

↳ End-to-End workflow explanation

↳ Formal conclusion of the research.

Scientific Method:-

→ To be termed scientific a method of inquery must be based on imperical and measurable evidence subject to specific principles of reasoning.

→ This scientific method attempt to minimize the influence of researcher bias on the outcome of an experiment.

→ The scientific method is the process by which we scientists, collectively and overtime, endable to construct and accurate (i.e reliable, consistent, and non arbitrary) representation of the world.

Auto M.L

Classical . M.L

↳ Supervised M.L

↳ unsupervised M.L

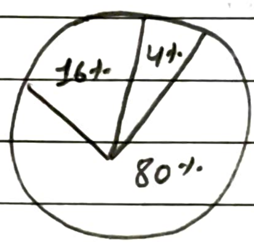
Automated ML

↳ In 2018

The 'why' and 'what'

1) The why: Pain points of traditional M.L

↳ The 80/20 Rule:



→ 80% Datacleaning

→ 16% Data collection

→ 4% Actual modeling

2) Deployment speed ("The Snail's pace")

40%	20%	14%	(company's)
6-months	3-months	7 days	(Duration for M.L projects)

3) Failure Rates ("The Valley of Death")

In 2018 Gartner predicted:-

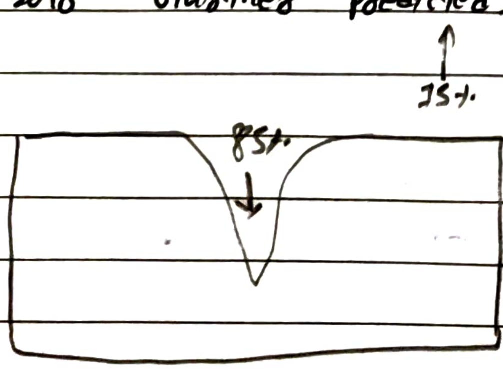


fig: AI Project valley of Death

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The "why" : understanding the Solution:

The "No free Lunch"
- No fixed algorithm for a single problem.

Introduction to AutoML
- process of automating the time-consuming iterative tasks

Core concept:-

AutoML replaces manual "trial & error"

i) Data preprocessing
ii) Feature Engineering
iii) Model selection
iv) Hyperparameter tuning

AutoML is not AI, it is a tool:-

- Define the business problem or objective.
- collect, label and curate the initial dataset.
- Interpret the ethical implications or bias within the result.

The benefits:-

- Democratization
- Efficiency.

The Result:

→ A shift from a "craftman" to an "industrial" approach.

Limitations :-
"Not a magic"

1) Black Box Nature
   ↳ why did it selected this algorithm (banking or health)

ii) computation cost :- Significant computation power

iii) Data Quality : cannot fix "garbage" data.
   ↳ Clean "empty data"
   ↳ Not efficient model

Step by step implementation of AutoML

Step 1: Data Loading & Integration,
* AutoML : PyCaret
• Load dataset = data = load_csv ('myfiles.csv')

Step 2 : Initialization
   ↳ Tell AutoML what to predict
   ↳ Pass setup function & specify the target column,

Step 3: Setup (data, target = 'column_name')

↳ Train dozen of algorithms using k-fold cross validation.
   It rank them on leaderboard.

best_model = compare_models()

Step4: Analysis & Diagnostics
↳ Don't trust blindly
↳ Generate plots to check for errors.

key checks:
↳ confusion matrix
↳ feature Importance to see which column is giving the result.

code: Plot_model (best_model, Plot = 'confusion_matrix')

Step 5:
Finalization & deploy
Action: Finalize the model & save it to a file like a ".Pkl" file.

code: save_model (best_model, 'my_final_ai')

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Program implementation:-

```
Pip install git+https://github.com/Pycaret/Pycaret.git@master
upgrade

# selection best model and confusion matrix interpretation
from Pycaret.classification import *
from Pycaret.datasets import get_data

dataset = get_data ("iris")
exp = setup (data= dataset, target = "species", session_id = 123)

best_model = compare_models()
print (f" Best model: {best_model}")

Plot_model (best_model, plot = 'confusion_matrix')

# New data given & checked the prediction

import pandas as pd
new_flower = pd. Dataframe ({'sepal_length': [5.1], 'sepal_width': [3.5]
'Petal_length': [1.4], 'Petal_width': [0.2]})

prediction = predict_model (best_model, data = new_flower)
print (prediction)
Save_model (best_model, "su")
```

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1) ~~Aptitude~~
2) technical MCQ → { C / J / P → DSA }
OS and Networking
3) RDBMS
4) Automata fix
5) Sudo code

3) competitive ~~coding~~ coding round
4) HR Technical round

---

1) $\boxed{\text{JDK}}$ 1.8
2) netbeans / eclipse

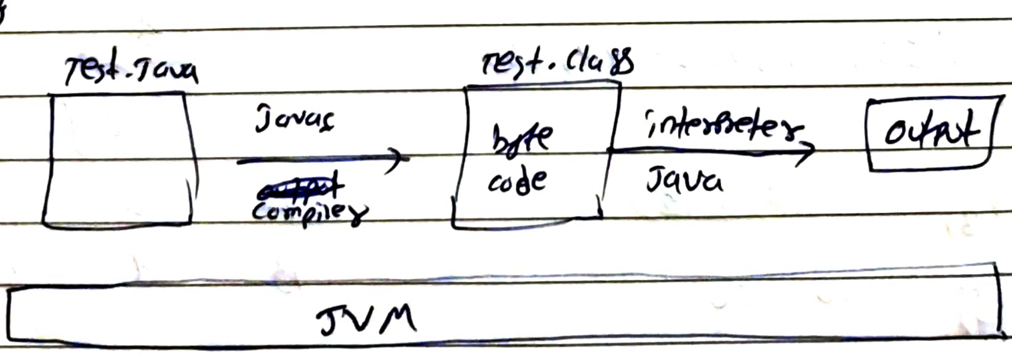
(1) Why Java is Platform independent.

Java
1) Object oriented programming language.
a) ~~data~~ class & objects
b) data encapsulation
c) data hiding
d) data abstraction
e) Polymorphism
f) Inheritance

{Java development kit}
JDK
JRE
↳ Java run time environment

TEST.Java

```
class TEST{
  public static void main (String args[]){
    System.out.println ("Hello world");}
}
```



Garbage collection

3) collect the obhet which no longer needed for program based on least used algo.
2) finalize()
3) destory()

Note: unmagned part by writing a program

Method to free memory
↳ gc / freememory()

Cmd:
folder / Javac Test.Java
run / Java Test

why public main function:
1) Java call main() from outside the program.

why static: Because Jvm call the main function using class name directly.

System.out.println(args[0]);

Data type preemptive : int, float.
wrapper class:

variable:

```
class TEST{
    public static void main (string args []){
    int i = 10;
    float k = 9.8;
    }
  }
}
```

{literals}

Primitive type   wrapper class   Literal   {strings, characters}

11
0b11
011
0x11

short   integer
byte
int
long   Double
float
double

Autoboxing & unboxing

int i = 10;
integer k = i; // Auto boxing

integer m1 = 90;
int m2 = m1 // unboxing

Diff betw type casting & conversion

```
class TEST {
    public static void main (String args[])
    {
      int i=10;
      Integer k= i; // Auto boxing
      integer m1 = 90;
      int m2 = m1; // unboxing

     float k1 = (float) 98 // type casting
     int k2 = integer.parseInt ("98");
     System.out.println (k1+k2);
```

1) javac TEST.java
2) java TEST

(Array)

1) collection of same type data
2) allocates memory contiguously
3) ordered collection
4) array represent by base element address.

int k[] = {11, 22, 33}

11	12	13

1000 1004

int a[1000];

0 1 2 ... 997 998 999

1000 t1 O(1) t2

t1 = t2 = constant time.

Disadvantage
- static

Advantage
- fast element access.

memory address = base address + index * byte size

for t2

1000 + 998 × 4 = ?

(1016)

A B

int [] a = {66, 11, 22, 23, 44, 55};

{55, 11, 22, 23, 44, 66}

19

3

4

~~B×2~~ =

~~B×1~~

A	3	10
B	4	15

(~~5×3=15~~)

(3)
A × 5 × 10 = 50
(4)
B × 1 × 15 = 15

65

~~A (3×5)×10~~ =

Q1) Josh went to the market of buy N apples, He found two shops, shop A and B, where apples were being sold in lots. He can buy any number of the complete lot(s) but not loose apples. He is confused with the price and wants you to figure out the minimum cost to buy exactly N apples. write an algorithm for Josh to calculate the minimum cost to buy exactly N apples:

Shop A → M1 apples per lot, price P1

Shop B → M2 apples per lot, price P2

we must: Buy exactly N apples, minimize total cost.

we try: • Take x lots from shop A

• Remaining from shop B

so :- $x * M1 + y * M2 = N$

we loop over possible x, and check if remaining apples can be exactly bought from the shop B

Sample input:

Total - 19

	no	price
ShopA -	3	10
ShopB -	4	15

Sample output:

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```
def min_cost (N, m1, P1, m2, P2):
    min_cost = float ('inf')

    # Try all possible lots from shop A
    for x in range (0, N // m1 + 1):
        apples_from_A = x * m1
        remaining = N - apples_from_A

        # check if remaining apples can be bought from shop B

        if remaining % m2 == 0:
            y = remaining // m2
            cost = x * P1 + y * P2
            min_cost = min (min_cost, cost)
    return min_cost if min_cost != float ('inf') else -1

#Input
N = int (input ())
m1, P1 = map(int, input().split())
m2, P2 = map (int, input(). split())

#output
print (min_cost (N, m1, P1, m2, P2))
```

```
class question1{
    public static void main (String args[])
    {
     int t= 19;
     int m1 =3;
     int P1 = 10;
     int m2 = 4;
     int P2 =15;
     int min_price = 9999;
     for (int i=1; i*m1<=t; i++)
    {
      int rem=t - i*m1;
      if (rem%4 ==0)
      { int rem = t - i*m1;
       if (rem%4 ==0){
          int lot = rem/4;
          int cost_price = i*P1 + lot*P2;
          if (cost_price < minprice
```

Right rotation : closewise

Left rotation : Anti-clockwise

| 11 | 4 | 9 | 15 | 45 |

temp = 45

```
import java.util.*;
Class Array {
  public static void main( string args[]){
```

Rotation ≠ Reverse

```
  int arr[] = {11,4,9,15,45}
  int n = arr.length;
  int k = 2;
  K = k%n;

  for (int i=0; i<k; i++){
    int last = arr[n-1];

    for (int j=n-1; j>0; j--){
      arr[j] = arr[j-1];
    }
    arr[0] = last;
  }
  system.out.println (Arrays.toString(arr));
 }
}
```

| 11 | 4 | 9 | 15 | 45 |

| 11 | 4 | 9 | 45 | 15 |

| 11 | 4 | 45 | 9 | 15 |

| 11 | 45 | 4 | 9 | 15 |

| 45 | 11 | 4 | 9 | 15 |

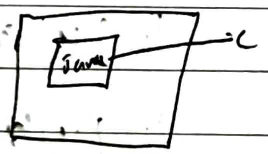
```
Class Reverse{
    public static void main( string args[]){
   int arr[] = {11,4,9,15,45};
   int s = arr.length;
   int temp = arr [s-1];
   for (int k = s-2; k>=0; k--)
    { arr[k+1] = arr[k];
    }
    arr[0] = temp;
    for (int m : arr) {
    system.out.println (m);
  }
  }
```

website NetBeans

string

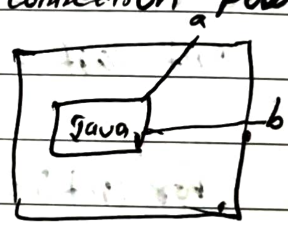
1) an array of characters
2) immutable - not update

```
class stringcompare{
  public static void main (string arg[]){
    string a = "Java";
    string b = "Ja" + "va";
    string c = new.string ("Java");
    system.out.println (a==b);
    system.out.println (a==c);
  }
}
```



Heap stack

String connection pool



where the object and variables are stored?

Characters in array of strings

Print all the ^ strings ~~in an array~~

```
class stringchar{
  public

    string str = "Hello";

    for (int i =0; i<= str.length (); i++) {
     system.out.println (str.charAt(i));
    }
  }
}
```

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write a function to find the factorial of a number.

```
~~public class classEx1{~~
    ~~public int fact (int~~ n){
public class Factorial{
// function
   public static int factorial (int n){
     int fact = 1;

     for (int i = 1; i <= n; i++){
        fact = fact * i;
     }
       return fact;
   }

public static void factorial (string[] args){

   int result = factorial (5);
   system.out.println (result);
  }
}
```

difference between instance & static function

pragma directives.

```
Public class ProgramEx2{
  public static void main (String arg[]){
    System.out.println ("I am from main");
  }
  static {
    System.out.println ("I am from static");
  }
```

// static method is initialized first before ~~static~~ main

Function overloading :-

Polymorphism
- static
- Dynamic

Function signature same but depends on
1) No of Parameters
2) data type of Parameter
3) sequence of Parameter

Note:- Never depends on return type of the function

```
Public class functionoverloading {
  Public void sum (int, a , int b){
    System.out.Println ("Result =" +(a+b));
  }
  Public void sum (int a, int b, int c){
    System.out.Println (" Result =", +(a+b+c));
  }
  } Public void sum (int a, double b) {
    System.out.Println ("Result =", + (a+b));
  } public static void main ( String args[]){
    Sum(1,2.3);
```

```
functionoverloading f= new functionoverloading();
    f.sum(11,23);
    f.sum(11,2);
}
}
```

* write a function for n number of inputs for use case.

```
Public class Arrayparam{
 Public int sum(int a[] ~~int b[]~~){

    int s=0;
    for(int i : a){
        s=s+i;
    }
    return s;
 }
 Public static void main(string args[]){

 Arrayparam m1=new Arrayparam();
  ~~m1.sum({11,22,33,44})~~
  int j[]={11,22,33,44,55};
  ~~m1.sum(a);~~
  system.out.println(m1.sum(j));

}
```

Variable length argument function

```
Public class variablelength {
  public static int sum (int... a){
    int s=0;
    for (int i : a){
      s=s+i;
    }
    return s;
  }
  Public static void main (String args[]){
    ~~System.out.Println (sum(1,2));~~

  System.out.Println (variablelength.sum(1,2));
  }
}
```

Input from user

Streams in Java

System.err

System.out

System.in

using

1) utility class - Scanner
   Java.util.Scanner

2) Standard input stream - BufferedReader
   Java.io.BufferedReader

1, 13, 25, 37
2, 14, 26, 38, 50, 62
3,

Class & object : store the properties and behaviour for different types of entity, we need to create a class

Students	S1	S2
name	name=aa	name=bb
email	email=a@.c	email=b@b.c
display()	display()	display()

Data hiding, encapsulation

constructor:

[:0] [:0] [stat .

getter & setter method (cognizant).

- Instance, class, object, static and all

Data Structure.

- If we store the data dynamically using different algorithms
Package name.

Java. util
→ (collection framework)

collection

List	set	map
1) collection	1) collection	1) store using key value Pair
2) ordered collection	2) unordered data	2) keys are unique
3) duplicate values allowed	3) No duplicates values	↳ value can repeat
4) Array list / Vector	4) mutable	3) No indexing

Generic class, interface, Template class

object: built in class lang Package
↓
user class
↓

// strongly type data
ArrayList<string> arr2 = new ArrayList<string> ();
↓
Java generic class

{infosys
cognitive}

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(iterator)

Pro: Store n number of student name and take input from user for any employee then search in the array list whether the employee name is avaible or not if available then remove the name and display all employee. name

Vector class (list implementation)

```
public class vectorEx{
    public static void main (String args[]){
        vector v1=new vector();          // capacity    size
        v1.add(11);
        v1.add(12);
        System.out.println("capacity="+v1.capacity());
        system.out.println("size="+v1.size());
        Vector v2=new Vector(3,4);
                     ↳ will increase size till 4 after 3
        v2.add(1);     fill up.
        v2.add(12);
        system    ,,    ("capacity
```

superclass
↑
object
each class ↗

.net

java.lang(object,

Day:4

```
import java.util.stack; import java.lang.object
public class DatastructureE1 (
    public static void main (String args[]) {
        stack s1 = new stack();
        stack <string> s2 new = stack();
         s1 push (11);
         s2. push ("aaa");
         s1. push ("98.9");
         s2. push ("11");
         system.out.println ("No of elements ="+ a2.size();
         system.out. println (" s2 contains ("ppp")); // False

         for (Object ; : s1{
             system.out. println(s); }
```

wap to store n number of students in java.

```
import. java.util.scanner

class student{
    string name;
    int rollno;

    student
```

Job, GATE

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wap to store n number of student name

```
import java.util.Scanner

public class studentname {
    public static void main (string args[]){

    Scanner Scanner = new Scanner (system.in)
    system.out.println ("Enter no of student");
    int n = next Int ( scanner.nextInt();
    string [] names = new string [n];

    for (int i=0; i<n; i++){
      system.out.println(" Enter no of students
```

Hash table

```
import java.util.Hashset;
  public class HashTableEx{
      public static void main (string args []) {
          Hashset. hs = new Hashset ();
           hs.add (11);
           hs.add (22);
           hs.add (23);
           hs.add (24);
           hs.add (11);
      system.out.println ("No of elements = " +hs.size());

    for (int i=0 ; i<n ; i++){
      system.out.println (" Elements :", (i)
    for (object o : hs){
      system.out.println (o);
    } }
```

~~Has~~ Has, is, relationship

Higher order function :-

multithreading

Iterator / Enumerator.

↓ can remove data

↓ can't remove data

~~Iterator itr = hs.iterator();~~

```
while (itr.hasNext())
{ ~~print~~
    System.out.println(itr.next());
  }
}
```

wap a to display no of unique elements in an array

Map in Java

1) Dynamic collection
2) Store the values using

{Java Swing}

10th

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```
import java.util.*;
class students{
    string name, email, mobile;
    public students (string nm, string em, string mob){
        This.name = nm;
        this.email = em;
        this.mobile = mob;
    }
    public void display (){
        system.out.println ("Name =" + this.name);
        system.out.println ("Email =" + this.email);
        system.out.println ("mobile =" + this.mobile);
    }
}
public class HashmapEx3{
    public static void main (string arg[])
{
    Hashmap <Integer, students > hmap = new Hashmap <integer,
    student

    students s1 = new students ("Raj, "r@gmail.com", "gsas");
    hmap.put (101, s1);

    students s2 = new students ("vipin", "v@gmail.com", "776553");
    hmap.put (102, s2);

    for (map.Entry m : hmap.entryset ()){
        system.out.println ("Roll" + m.getkey());

    students s = (students) m.getvalue();
    s display ();
}}
```

mern, react, node, angular, Django

Java

.dotnet

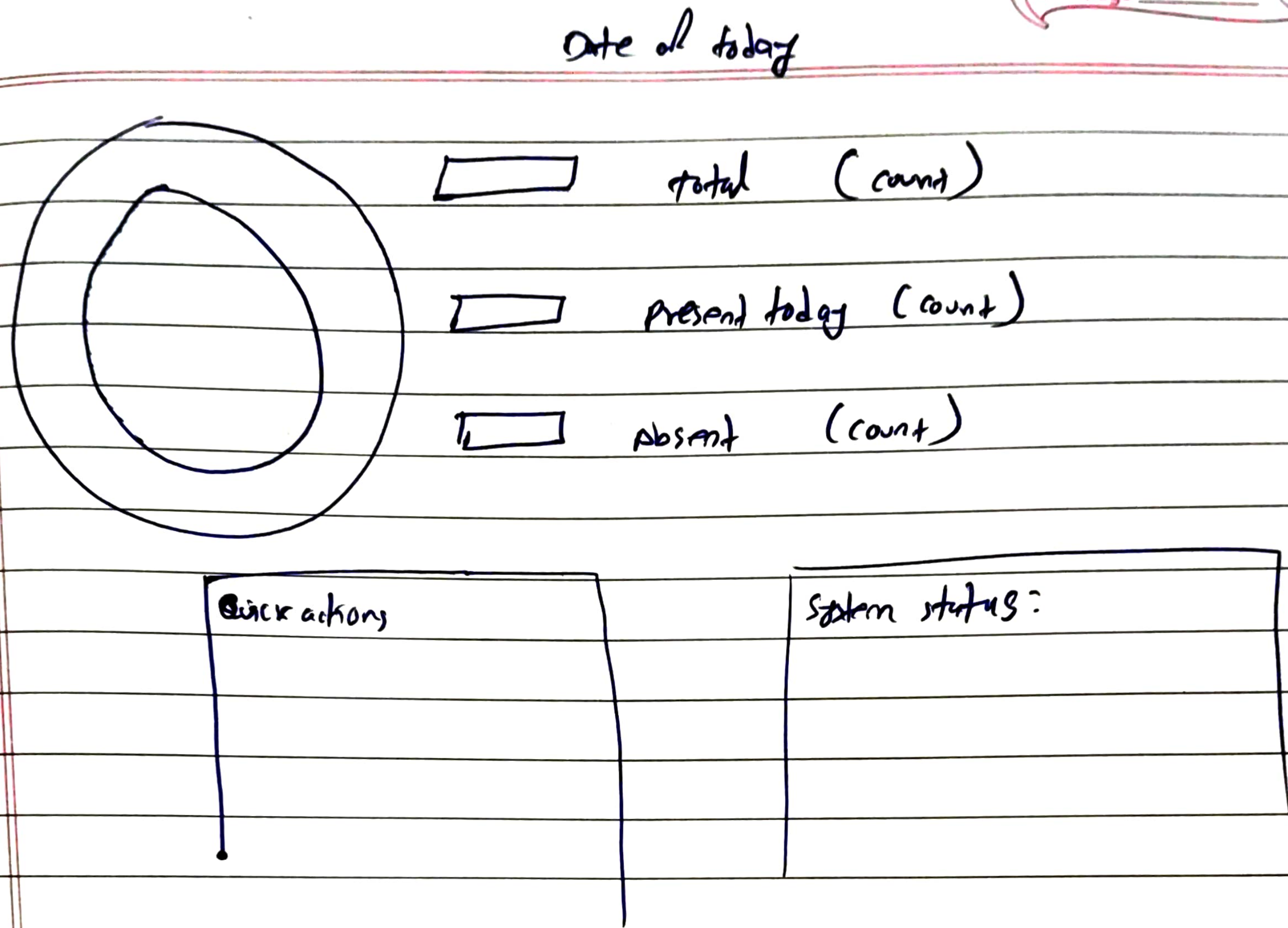
flutter

visualisation of project.

Cloud computing

x Project description

- Project details ~~Introduction~~
- Acknowledgement
- Abstract
- Introduction
- DSA and algorithm used
- methodology (steps to complete)
- case (output + screenshot)
- future
- limitation
- conclusion



Nav bar hover improvement (where we are currently)

Filters in students page in (Admin) → Absent / Present
Also in leaves Approved / Pending / Rejected
Also in Records Search option (id)

Admin block

Student image proper rectangle card
with visual graph or diagram of % of attendance
hover effect on nav. where we currently are.

Student block